GEONSIK MOON

(917) 257-7860 | geonsik.moon@columbia.edu | linkedin.com/in/gsmoon97 | github.com/gsmoon97

Education

Columbia University — Expected Dec 2026
Master of Science, Computer Science (ML Track), GPA: 4.00/4.00 — *New York, NY*

- Teaching Assistant; High Performance Machine Learning (profiling, CUDA, quantization, FlashAttention)
- Research Assistant; built retrieval-grounded (RAG) LLM pipelines to classify green-energy legislation
- IBM Research Mentored Project: integrated Granite Speech 3.3 into IBM's Foundation Model Stack (FMS) with a Conformer encoder, Q-Former projector, LoRA-adapted decoder, and `torch.compile` optimization

National University of Singapore — Jun 2022
Bachelor of Computing, Computer Science (Honors) — *Singapore*

- Specialization Awards: Distinction in Artificial Intelligence, Merit in Database Systems

Experience

AI Platform Developer Intern — May 2026 – Present
IBM Research — *Yorktown Heights, NY*

- Built open-source ML compiler infrastructure for *torch-spyre*, IBM's PyTorch backend for the Spyre AI inference accelerator, extending the `torch.compile` Inductor lowering stack that generates optimized hardware kernels
- Designed a provenance API that links on-device profiler events back to the originating PyTorch source line and ATen operator, enabling source-level attribution of kernel hotspots to guide inference performance optimization

LLM Training Operation Specialist — May 2024 – Aug 2025
ByteDance Seed — *Singapore*

- Co-authored *AetherCode*, an open-source competitive programming benchmark released on Hugging Face
- Orchestrated end-to-end data pipelines for the *CodeContests+* RL training benchmark, supervising ∼70 annotators across 17K+ training samples to align multi-agent code-generation systems
- Engineered the evaluation feedback loop for software-engineering agents, diagnosing failure modes across 3K+ outputs and driving a 15% improvement in task-completion performance

NLP Research Assistant — Sep 2022 – Feb 2024
National University of Singapore — *Singapore*

- Investigated encoder-only vs. decoder-only LLM architectures for lexical semantic understanding and built an incremental LLM-based framework for timeline summarization, resulting in two top-tier conference papers
- Designed two full-stack, web-based grammatical error correction systems with Docker and Flask
- Fine-tuned open-source LLMs (Mistral, Llama, FLAN-T5) with QLoRA (4-bit quantization) for parameter-efficient training, improving convergence stability and systematically tracking experiments with Weights & Biases

Machine Learning Engineer — May 2022 – Sep 2022
Apple (via TransPerfect) — *Singapore*

- Enhanced Siri's Natural Language Understanding through large-scale error analysis of production interaction logs
- Built automated dialogue-data optimization pipelines using transformer-based synthetic data augmentation (BERT, T5), reducing dataset noise by ∼30% and improving downstream response relevance

Publications

[1] *AetherCode: Evaluating LLMs' Ability to Win in Premier Programming Competitions.* ICLR 2026.
[2] *Are Decoder-Only LMs Better than Encoder-Only LMs in Understanding Word Meaning?.* ACL 2024.
[3] *From Moments to Milestones: Incremental Timeline Summarization Leveraging Large Language Models.* ACL 2024.
[4] *WAMP: Writing, Annotation, and Marking Platform.* IJCNLP-AACL 2023 (System Demonstrations).
[5] *ALLECS: A Lightweight Language Error Correction System.* EACL 2023 (System Demonstrations).

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, SQL
ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn
ML Systems & Performance: CUDA, PyTorch Compilation (Dynamo, AOTAutograd, Inductor), Profiling
LLM Training & Evaluation: LoRA/PEFT, QLoRA, LM Eval Harness, Weights & Biases
LLM Infrastructure: LangChain/LangGraph, vLLM, Ollama, llama.cpp, Chroma, Pinecone
Systems & DevOps: Docker, Kubernetes, AWS (Bedrock, EC2, S3), GCP, Git